

TESLIN RIVER WATERSHED
COMMUNITY STEWARDSHIP PROGRAM, 2005

The Yukon River Panel
Restoration and Enhancement Fund
CRE-47-05

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ABSTRACT

The Teslin Tlingit Council continued their stewardship of the Teslin River Watershed in 2005. A total of 9 streams was walked and obstructions to upstream migration of juvenile chinook salmon located. Beaver dams and/or logjam obstructions were observed in 4 of these streams including: Dave, Grayling, Brooks Brook and Sterlin Creeks. Beaver dam and logjam obstructions were breached following the Department of Fisheries and Oceans (DFO) Guidelines. A logjam associated with an old beaver dam observed near the mouth of the Swift River (north) during an over flight was thought to be an obstruction to migrating chinook spawners and was later breached by the stewardship crew. Minnow traps were set in 10 streams investigated to determine utilisation by juvenile chinook salmon. Stream habitat and fisheries assessments were conducted on tributaries entering Morley, Quiet and Sandy lakes. No juvenile chinook salmon were captured in minnow traps set in these tributaries. A total of 17 DNA samples was collected from post-spawned chinook in the Big Salmon River. An attempt was also made to collect DNA samples from post spawned chinook in the lower Teslin River but no chinook carcasses were found at the time of the survey.

TABLE OF CONTENTS

ABSTRACT	i
List of Figures	iii
List of Appendices	iii
INTRODUCTION	1
STUDY AREA	1
METHODS:	3
Pre-season planning	3
Field methods.....	3
Salmon Restoration & Enhancement plan.....	4
RESULTS.....	4
DISCUSSION.....	5
ACKNOWLEDGMENTS.....	5
REFERENCES:	5

List of Figures

Figure 1. Study Area -- Major Tributaries of the Teslin River Drainage.....	2
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List of Appendices

Appendix 1. Work Plan for Stewardship Activities in 2005.....	6
Appendix 2. Teslin Tlingit Council Traditional Territory, Teslin Watershed Salmon Restoration and Enhancement Plan.....	7
Appendix 3. Obstruction Removal and Maintenance of Upstream Migration for Salmon Field Sheets.....	26
Appendix 4. Stream Habitat Assessment Field Data Sheets.....	35
Appendix 5. Juvenile Chinook Salmon Sampling Data Sheets.....	41

INTRODUCTION

The Teslin Tlingit Council (TTC) in cooperation with federal and other agencies has conducted a Community Salmon Stewardship program annually since 2003 (Wilson, 2004, 2005). This program was developed as a long-term initiative to involve community members in efforts to conserve and restore wild salmon stocks and the habitats on which they depend in the Teslin River Watershed.

Stewardship activities and recommendations are compiled each year, by stream or lake, and entered in the Teslin Tlingit Council, Traditional Territory, Teslin Watershed Restoration and Enhancement Plan. Annual work plans are developed in consideration of work that has been conducted in the past, and of the recommendations made. The primary goal of the plan is to restore salmon stocks and habitats in the Teslin River Watershed. The secondary goal of the plan is to enhance salmon stocks and habitats in the Teslin River Watershed.

This report provides a summary of stewardship activities in 2005 and recommendations in the Teslin Watershed Restoration and Enhancement Plan. The specific objectives of the stewardship component are as follows:

- As a first priority, ensure that salmon habitats are accessible to returning adult salmon;
- Monitor juvenile salmon populations and habitats; and
- Assist managers in the collection of biological or other samples.

A key component of this program is to provide training for Teslin Tlingit Council members to build capacity within the community for effective implementation of stewardship initiatives. Field technicians involved in stewardship activities must have received certification in the 'Yukon Fisheries Field Technician Course' offered by Yukon College. On-going field training is provided by DFO personnel and other agencies retained by TTC. Technicians also attend and receive certification in applicable safety courses.

The Yukon River Panel Salmon Restoration & Enhancement Fund under the U.S./Canada Yukon River Agreement of the Pacific Salmon Treaty and the Teslin Tlingit Council Training Fund provided funding for this program.

STUDY AREA

The Teslin River Watershed (Fig. 1) is one of 6 major watersheds identified in the Canadian portion of the Yukon River drainage basin. The watershed is primarily located within the traditional lands of the Teslin Tlingit Council and straddles the Yukon-British Columbia border. Chinook salmon are known to spawn in 24 streams and rivers in the watershed, with chum salmon utilising a smaller number of streams. The river downstream of Teslin Lake is known as the "Lower Teslin River" and the section upstream of Teslin Lake is the "Upper Teslin River".

METHODS:

Pre-season planning

TTC Lands and Resources representatives and J. Wilson & Associates met on April 11 with DFO technical advisors to discuss plans for the 2005 field season. Management concerns and data gaps relating to salmon bearing streams in the drainage were identified with guidance from the Teslin Tlingit Restoration & Enhancement Plan and then evaluated as to their importance and economic feasibility. Projects were given priority and incorporated into a work plan. (see work plan Appendix 1). This plan was intended as a guide for scheduling work activities while allowing for some flexibility depending on environmental conditions.

In June and early July, stewardship technicians received certification in field safety courses held in Teslin and Whitehorse. These included Wilderness First Aid, Swift Water Rescue Training and firearms safety.

Field methods

Three technicians from the community of Teslin, all of who had successfully completed the Yukon Fisheries Field Technician Course, conducted field activities outlined in the work plan. The technicians recorded activities conducted in the field on field forms that were developed in accordance with the Yukon River Panel data reporting protocols. At the end of each week, completed forms were photocopied and the information entered into a computer.

Obstruction removal and maintenance

Streams and/or rivers were investigated on foot, by boat or by floatplane to locate possible obstructions to upstream migrating salmon. Obstruction assessment and mitigation was conducted in accordance with the "*Yukon River Panel Protocol for Obstruction Management in Canadian YR R&E Projects*" (2005).

Juvenile salmon sampling

Gee-type minnow traps, baited with cat food, were used to capture jcs in streams. A minimum subset of 30 jcs per sampling station was measured for fork length to the nearest millimeter and weighed to the nearest 0.1 gram. All juvenile trapping and reporting procedures abided by the "*Yukon River Panel Protocol for Collection and Reporting of Data from Juvenile Salmon Sampled in Canadian R&E Projects*" (2004).

DNA sample collection

Adult chinook salmon were captured by gill netting and/or by angling. A small portion of tissue was removed from the operculum using a paper punch. All fish were released immediately after sampling. Tissue samples were preserved in alcohol and sent to Pat Milligan, DFO stock assessment biologist, for DNA analysis.

Stream habitat assessment

Stream habitat assessments were conducted in accordance with DFO 'Streamkeepers' stream assessment procedures using equipment supplied in the 'Streamkeepers' kit.

Salmon Restoration & Enhancement plan

Stewardship activities conducted during the season were compiled and entered in the Teslin River Watershed Restoration & Enhancement plan. Major tributaries are listed in order in relation to the Lower Teslin River from the furthest downstream to the furthest upstream. New streams and lakes are added as required. A brief description of each stream or lake is given. Recommendations for follow-up activities are made where applicable.

RESULTS

A total of 9 streams was walked to locate obstructions to juvenile chinook salmon migration. Beaver dams and/or logjam obstructions were observed and breached in 4 of the streams investigated including: Dave, Grayling, Brooks Brook and Sterlin creeks. Of these, Brooks Brook, Grayling and Sterlin creeks were re-visited to monitor beaver activity and jcs utilisation. Dave Creek was not re-visited due to time constraints. The breached beaver dams in Brooks Brook had remained open and juvenile fish were observed in reaches above them. The beaver dam in Grayling Creek was very high (approximately 3 m) and had been rebuilt. The crew did not breach this dam a second time, as the beavers were still active in this area. There was very little water flowing beneath this dam and no juvenile fish were observed in reaches downstream.

During an over flight of the Swift River (north), a chinook spawning river, a logjam associated with an old beaver dam was observed near the mouth and was thought to be an obstruction to migrating chinook spawners. The mouth of the river was accessed by boat on July 20 and the obstruction breached. Completed obstruction removal and maintenance of upstream migration for salmon field sheets are given in Appendix 3.

There were no jcs captured in minnow traps set in tributaries of Morley, Quiet and Sandy lakes. The streams entering Morley Lake were small and appeared to have limited capability of supporting rearing jcs. Streams entering Quiet and Sandy lakes appeared to have suitable rearing habitat for jcs although none were captured in minnow traps that were set. Completed habitat and fisheries assessment sheets are given in Appendix 4.

Minnow traps were set in 10 streams to determine utilisation by juvenile chinook salmon. Jcs catches and sampling results are given in Appendix 5.

A total of 17 DNA samples was collected from post-spawned chinook in the Big Salmon River. An attempt was also made to collect DNA samples from post spawned chinook in the lower Teslin River but no chinook carcasses were found at the time of the survey.

DISCUSSION

The work schedule outlined for stewardship activities in the Teslin River Watershed in 2005 was generally adhered to. There were no instances when weather conditions prevented a project from being undertaken. There was, however, some difficulty experienced in daily organization and efficient use of time. This resulted in fewer streams being investigated than expected and inadequate follow up in some cases. It was apparent in some instances that the procedures outlined in the Yukon River Panel protocols were not clearly understood by the crew. Transportation was also an issue at times when the only vehicle available for the crew was in use by other TTC staff.

In future stewardship programs by TTC the following recommendations should be considered:

- A vehicle should be available specifically for the stewardship crew during the field season.
- DFO and others agencies retained by TTC should continue to provide in-field training.
- The contractor retained by TTC and the stewardship crew should develop a schedule of daily activities and contingency plans at the beginning of each week.
- field technicians should continue to attend and receive certification in applicable safety courses.

ACKNOWLEDGMENTS

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Yukon River Panel. 2005. Protocol for Collection and Reporting of Data from Juvenile Salmon Sampled in Canadian R&E Projects.

Wilson J. 2004. Teslin River Sub-Basin Community Stewardship Program, 2003. Prepared for the Yukon River Panel, Project CRE-47-03.

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Appendix 1. Work Plan for Teslin River Watershed Community Stewardship Program, 2005.

DATE	ACTIVITY	
April	Development of workplan and application for required permits	Whitehorse
May – June	Fisheries technician safety and other training/certification	
July 4 - 15	Obstruction management in rearing streams. Juvenile chinook monitoring.	Day trips from Teslin
July 18 - 22	Obstruction assessment in Swift River (north) including over-flight and mitigation as required. Monitoring juvenile salmon populations and obstruction management in rearing streams of the lower Teslin River.	Camp out
July 25 - 29	Stream habitat assessments of tributaries to Quiet Lake.	Camp out
August 1 - 5	Stream habitat assessments of tributaries to Morley Lake.	Day trips from Teslin
August 8 - 12	Continuation and follow up on obstruction management in rearing and spawning streams. Continuation of juvenile salmon monitoring programs. Conduct stock assessment projects as identified in season.	Day trips
August 15 - 19	Carcass sampling on Big Salmon River (ASL data and genetic material etc.)	Camp out
August 22 - 26	Carcass sampling on Teslin River (ASL data and genetic material etc.)	Day trips from Teslin

Appendix 2.

TESLIN TLINGIT COUNCIL TRADITIONAL TERRITORY TESLIN WATERSHED SALMON RESTORATION AND ENHANCEMENT PLAN (including portions of the Yukon River Mid – Mainstem Watershed).

LOWER TESLIN RIVER

The lower Teslin River drains Teslin Lake and flows in a north-westerly direction for 192 km to its confluence with the Yukon River. Chinook spawning occurs in a number of areas, with the best known and most stable at the “rapids” approximately 3 km downstream of Johnson’s crossing. Chum spawning occurs in at least two locations, both some distance downstream from Johnson’s Crossing. Access to the river is gained by boat from Johnson’s Crossing or by a road that leads to the bank above the rapids.

2005

August 25-26– surveyed river by boat to obtain DNA samples from post-spawned chinook. Looked for carcasses and angled for live post-spawned chinook between Johnson’s Crossing and the confluence of Boswell Creek. No chinook carcasses were observed and angling captured no chinook. High water and turbid conditions hampered the ability to see fish and spawning areas.

Recommendations for 2006

Investigate other means of obtaining DNA samples from chinook salmon in the Lower Teslin River.

Tributaries:

Dave Creek – 8703727

drainage area: 70 km²

Dave Creek enters the Teslin River from the east approximately 29 km downstream of Johnson's Crossing. Chinook salmon rear in the creek. Beavers may dam the creek. Access to the creek is by boat.

2005

July 20 – surveyed lower 200 m of creek on foot. Breached active beaver dam located at the mouth of creek. No minnow traps were set.

Recommendations for 2006

Check stream for beaver activity. Monitor utilisation by jcs.

Hundred Mile Creek –

drainage area: 116 km²

Hundred Mile Creek enters the Teslin River from the east approximately 35 km downstream of Johnson's Crossing. Chinook salmon rear in the creek. Beavers may dam the creek. Access to the creek is by boat.

2005

July 20-21 – Surveyed lower 200 m of creek on foot. No obstructions observed. Set 5 minnow traps. Captured and sampled 12 jcs.

Recommendations for 2006

Check stream for beaver activity. Monitor utilisation by jcs.

Swift River - 8703271

(note: there is another Swift River, which is tributary to Teslin Lake)

drainage area: 256 km²

The Swift River enters the Teslin River from the east approximately 47 km downstream of Johnson's Crossing. Chinook salmon spawn and rear in the river. Adults may have difficulty entering the river at low flows. Beaver may dam the river, particularly when flows are low. Access to the river is by boat.

2003

June 8 – flew creek. No beaver dams or obstructions observed.

Recommendations for 2004

Check stream for beaver activity. Collect DNA samples

2004

No over-flight was conducted. Local floatplanes were not available in late July as a result of extensive forest fire activity in the area.

Recommendations for 2005

Check stream for beaver activity. Collect DNA samples

2005

July 19 – flew creek. Observed one old beaver dam/logjam near the mouth of the river.

July 20 – went by boat to mouth of river. Manually removed woody debris that had collected on old beaver dam [above].

July 20-21 – set 5 minnow traps in the vicinity of the beaver dam/logjam. Captured and sampled 8 jcs.

Recommendations for 2006

Check stream for beaver activity. Collect DNA samples if feasible.

TESLIN LAKE

Teslin Lake - 870

Teslin Lake is a major headwater lake of the Yukon River and the largest lake in the Teslin River drainage. The lake lies across the Yukon-British Columbia border. It is 126 km long, averages 3.2 km in width and has an average depth of 59 meters. The Alaska Highway follows the northeast lakeshore and crosses an arm of the lake, Nisutlin Bay, near the town of Teslin.

2004

July 8 - beach seined at outlet of Teslin Lake (Teslin River) along west bank. Captured a total of 28 jcs from 3 sets. Other fish species captured included 153 juvenile whitefish, 6 arctic grayling and 3 slimy sculpins. Sampled subset of the total jcs captured.

July 13 - beach seined shoreline at mouth of Grayling Creek. Captured a total of 285 jcs. Sampled subset of the total jcs captured.

July 19 & 20 - distributed fishing licenses, buoys, catch calendars and weekly DFO salmon fishery updates to Teslin Tlingit members fishing in Teslin Lake and Teslin River.

Oct - Nov - set gillnets for chum salmon. Two chum salmon were captured in the Teslin Tlingit food fishery; one on November 3rd near the outlet of 10 Mile Creek and the other near the village of Teslin on November 17. Took samples for DNA analysis.

Recommendations for 2005

Conduct a mark-recapture study to determine if jcs migrate from the Teslin River to tributaries of Teslin Lake.

2005

A mark-recapture study was deferred and not identified as a priority project in the 2005 work plan.

Recommendations for 2006

Determine methods for a mark-recapture study in consultation with DFO personnel. If feasible, implement a mark-recapture study to determine if jcs migrate from the Teslin River to tributaries of Teslin Lake.

Tributaries:

Beaver Creek #1 -

Beaver Creek #1 enters Teslin Lake along the east shore approximately 7 km south of the village of Teslin. Nothing is known about chinook utilisation in this creek. Access is from the Alaska Highway crossing near the mouth. Beavers may dam the creek.

2005

July 27 - Surveyed creek on foot between the mouth and 500 m upstream of highway. No obstructions observed. Observed numerous unidentified fish (no traps set).

Recommendations for 2006

Check stream for beaver activity. Check for jcs utilisation

Beaver Creek #2 -

Beaver Creek #2 enters Teslin Lake along the east shore approximately 9 km south of the village of Teslin. Nothing is known about chinook utilisation in this creek. Access is from the Alaska Highway crossing near the mouth. Beavers may dam the creek.

2005

July 27 - Surveyed creek on foot between the mouth and 500 m upstream of highway. No obstructions observed. Observed numerous unidentified fish (no traps set).

Recommendations for 2006

Check stream for beaver activity. Check for jcs utilisation

Brooks-Brook (Gantiyakw)– 8704665

drainage area: 60 km²

Brooks-Brook (Tlingit name 'Gantiyakw') enters Teslin Lake along the east shore approximately 38 km north of the village of Teslin. Chinook salmon rear in the creek. The Alaska Highway crosses the creek near the mouth. The highway culvert is a barrier to small fish during high and low flow conditions. Beavers may dam the creek.

2003

June 23 -- Surveyed creek on foot. Saw active beaver dam 200 meters upstream of mouth. Breached dam. Saw fresh beaver tracks.

July 15-16 -- set minnow traps above and below culvert. Captured jcs below culvert but none above. Saw adult grayling above culvert.

Recommendations for 2004

Check stream for beaver activity. Monitor utilisation by jcs. Assess culvert.

2004

June 8 - set 5 minnow traps below highway culvert. Captured one over-wintering jcs (1+).

July 15 - set 4 minnow traps below highway culvert (Captured 1 jcs). Set 5 minnow traps above highway culvert (Captured 2 jcs). Sampled all jcs captured.

Recommendations for 2005

Check stream for beaver activity. Monitor utilisation by jcs.

2005

July 12-13 – surveyed creek on foot. Saw two beaver dams between mouth and highway culvert. The first beaver dam upstream of the mouth (BD-1) was older/inactive and located approximately 250 m downstream of the highway culvert. The second dam (BD-2) was active and located approximately 50 m upstream of BD-1. Set 6 minnow traps above BD-2 and 5 below BD-1. Captured one jcs above BD-2 and 14 jcs below BD-1. Sampled all jcs captured. Breached both beaver dams.

July 14- revisited creek to monitor beaver activity. Observed that breached dams had remained open. Observed unidentified juvenile fish above dams.

August 9-10 - set 3 minnow traps below breached beaver dams [above] and 5 above. Captured 8 jcs below dams and none above. Sampled all jcs captured.

Recommendations for 2006

Check stream for beaver activity. Monitor utilisation by jcs.

Deadman Creek – 8704909

drainage area: 150 km²

Deadman Creek enters Teslin Lake along the east shore approximately 27 km north of the village of Teslin. Chinook salmon rear in the creek. Anecdotal information suggests previous utilisation by chinook spawners. Access is from the Alaska Highway crossing near the mouth. Beavers may dam the creek. The creek is a good candidate for community/education enhancement projects.

2003

June 24 -- Survey creek on foot. No active beaver dams observed in lower reaches.

July 9-10 --Set minnow traps. No juvenile chinook captured at this time.

Recommendations for 2004

check stream for beaver activity. Monitor jcs utilisation. Initiate riparian planting project. Investigate methods to reintroduce chinook spawning population.

2004

June 17 - Survey creek on foot. No active beaver dams observed between highway bridge and 1 km upstream. Observed a logjam approximately 400 meters upstream of bridge, however, it did not appear to be a barrier.

July 14 - set 5 minnow traps between highway bridge and logjam located 400 m upstream of bridge (captured 7 jcs). Set 4 minnow traps above same logjam (captured 19 jcs). Sampled all jcs captured.

Recommendations for 2005

check stream for beaver activity. Monitor jcs utilisation. Initiate riparian planting project. Investigate methods to reintroduce chinook spawning population.

2005

Riparian planting project was not identified as a priority in the 2005 work plan.
Investigations of methods to reintroduce a chinook spawning population was not initiated.
July 5 - Surveyed creek on foot between the mouth and approx. 500 m upstream of highway bridge. No active beaver dams observed. Set 5 minnow traps throughout area surveyed.
Captured and sampled 2 jcs.
August 9-10 - set 10 minnow traps between mouth and 500 m upstream of highway bridge.
Captured and sampled 11 jcs.
Recommendations for 2006
check stream for beaver activity. Monitor jcs utilisation. Initiate riparian planting project.
Investigate methods to reintroduce chinook spawning population.

Lone Tree Creek – 8705099

drainage area: 41 km²

Twelve Mile creek enters Teslin Lake along the east shore approximately 10 km north of the village of Teslin. Rearing potential is limited. The creek has a steep gradient. Access is from the Alaska Highway crossing near the mouth.

2003

June 24 - Surveyed creek on foot. Observed very little good rearing habitat.

Recommendations for 2004

Check for utilisation by jcs.

2004

June 17 - surveyed creek from highway downstream to mouth. No obstructions observed.

July 6 - set 5 minnow traps below highway culvert (no jcs captured). Set 4 minnow traps above highway culvert (no jcs captured).

Recommendations for 2005

Check for jcs utilisation.

2005

No survey conducted due to time constraints.

Recommendations for 2006

Efforts should be made to monitor jcs utilisation.

Ten Mile Creek – 8705187

drainage area: 33 km²

Ten Mile Creek enters Teslin Lake along the east shore approximately 13 km north of the village of Teslin. Chinook rear in the creek. Access is from the Alaska Highway crossing near the mouth. Beavers may dam the creek.

2003

June 24 - Survey creek on foot. Logjam/old beaver dam complex located 200 meters upstream of lake. Took photos. Breached obstruction. Good rearing habitat observed. Culvert not likely a barrier to fish.

July 15-16 - set minnow traps. No jcs captured at this time.

Recommendations for 2004

Check stream for beaver activity. Monitor jcs utilisation

2004

June 16 - surveyed creek on foot. No obstructions observed between mouth and highway culvert and immediately upstream of culvert.

July 5 - set 5 minnow traps below highway culvert (no jcs captured). Set 4 minnow traps above highway culvert (no jcs captured).

Recommendations for 2005

Monitor jcs utilisation.

2005

No survey conducted due to time constraints.

Recommendations for 2006

Efforts should be made to monitor jcs utilisation.

Fox Creek – 8705359

drainage area: 56 km²

Fox Creek enters Teslin Lake along the east shore approximately 4 km north of the village of Teslin. Chinook salmon rear in the creek. The creek is susceptible to current and future development. The town sewage lagoons are located near the creek upstream of the Alaska Highway. The highway culvert may restrict fish passage during low flows. Access is by the Alaska Highway crossing near the mouth or by the entrance road to sewage lagoons. Beaver may dam the creek.

2003

June 24 - Surveyed creek on foot. No beaver dams observed.

July 8-9 - Conducted stream assessment in lower reaches. Took photos. Took water sample and sent to Norwest Labs for analysis. Collected benthic sample and sent to Bonnie Burns (Laberge Environmental Services) for identification. Set minnow traps. No jcs captured at this time.

Recommendations for 2004

initiate riparian planting project. Monitor jcs utilisation. Initiate culvert passage mitigation.

2004

June 16 - surveyed creek on foot. No obstructions observed between mouth and highway culvert and immediately upstream of culvert.

Recommendations for 2005

initiate riparian planting project. Monitor jcs utilisation.

2005

No survey conducted due to time constraints. The riparian planting project was not identified as a priority project in the 2005 work plan.

Recommendations for 2006

Efforts should be made to monitor jcs utilisation. Initiate a riparian planting project

Grayling Creek -

drainage area: 83 km²

Grayling Creek enters Teslin Lake along the west shore. Chinook salmon rear in the creek. Beavers may dam the creek. Access is by boat from Teslin Lake.

2003

July 2 - Saw a number of beaver dams in the lower reaches of creek and a considerable amount of large debris throughout. Took photo

Recommendations for 2004

Check stream for beaver activity. Monitor jcs utilisation.

2004

July 13 - beach seined along shore at mouth of creek. Captured a total of 285 jcs in 2 sets. Sampled subset of the total jcs captured.

Recommendations for 2005

Conduct mark-recapture study to determine origin of jcs. Check stream for beaver activity. Monitor jcs utilisation.

2005

August 11 – surveyed lower 250 m of creek on foot. Observed high active beaver dam (at least 3 m high) approximately 250 m upstream of mouth (BD-1). This dam held back a considerable amount of water and resulted in reduced flows downstream. Breached dam.

Recommendations for 2006

Check stream for beaver activity. Monitor jcs utilisation.

Fat (Flat) Creek – 8705232

drainage area: 800 km²

This creek is referred to as Fat Creek on some topographic maps but is known locally as Flat Creek. The creek enters Teslin Lake along the west shore. Chinook salmon rear in the creek. Beavers may dam the creek. Access is by boat from Teslin Lake.

2003

July 2 - saw large beaver dam in the lower reaches of creek. Took photos.

October 23 - conducted aerial survey of lower reaches with DFO personnel to investigate chum salmon spawning potential. No chum spawners observed.

Recommendations for 2004

Check stream for beaver activity. Monitor jcs utilisation.

2004

June 28 - survey lower reaches of creek by boat and on foot. No obstructions observed.

Recommendations for 2005

Check stream for beaver activity. Monitor jcs utilisation.

2005

No survey conducted due to time constraints.

Recommendations for 2006

Efforts should be made to check stream for beaver activity and to monitor jcs utilisation.

Sterlin Creek – 8705232022

drainage area: 81 km²

Sterlin Creek is a tributary of Flat Creek. Chinook salmon rear in the creek. Beavers may dam the creek. Access is by boat from Teslin Lake.

2003

July 2-3 - Saw 3 active beaver dams in lower reach. Took photos. Unidentified small fish observed below beaver dams.

Recommendations for 2004

Check stream for beaver activity. Monitor jcs utilisation.

2004

June 28 - survey creek on foot. Active beaver dam located approximately 150 meters upstream from Flat Creek confluence. Took photos of dam. No minnow traps were set but did see unidentified adult fish above the dam.

July 13 - set 5 minnow traps below beaver dam [above](captured 8 jcs). Set 4 minnow traps above this beaver dam (captured 5 jcs). Observed side channel around this dam that likely provided access for juvenile fish. Sampled jcs captured.

Recommendations for 2005

Check stream for beaver activity. Monitor jcs utilisation.

2005

July 4 - surveyed lower 100 m of creek. Beaver dam observed approximately 100 m upstream of mouth. Observed adult grayling above dam and adult grayling and unidentified juvenile fish below. No action taken at this time.

July 27 – surveyed creek 200 m upstream. Beaver dam located approximately 100 m upstream was washed out. Observed large beaver dam approximately 200 m upstream of mouth. Unidentified juvenile fish were observed below beaver dam and adult grayling above dam. Breached beaver dam.

Recommendations for 2006

Check stream for beaver activity and monitor jcs utilisation.

NISUTLIN RIVER

Nisutlin River – 874

drainage area: 11,984 km²

The Nisutlin River enters Nisutlin Bay approximately 11 km from the town of Teslin. Chinook salmon spawn and rear in the river. Access to the river is by boat from Teslin, by vehicle from the Canol Road at Sidney Lake, or by boat from the Canol crossing of Rose River #1.

2003

August 23 - Conducted aerial survey with DFO personnel to enumerate chinook spawners in index area.

October 23 - Conducted aerial survey between mouth and Thirtymile creek confluence with DFO personnel to investigate chum salmon spawning potential. No chum spawners observed at this time.

Recommendations for 2004

continue to participate in aerial enumeration surveys. Collect DNA samples.

Check reaches above Thirtymile creek confluence for chum spawning potential.

2004

August 18 - Aerial chinook enumeration survey of index area conducted by DFO. A total of 330 chinook spawners was counted.

October 7 - set gillnet for chum salmon at 'Crowknife Point' located approximately 2.3 km upstream of the mouth. Captured one whitefish but no chum salmon in one hour set.

Recommendations for 2005

participate in aerial enumeration surveys. Collect DNA samples. Check reaches above Thirtymile Creek confluence for chum spawning potential.

2005

DNA sample collection was deferred to accommodate other DNA collection projects in the Teslin River Watershed.

August 15 - Aerial chinook enumeration survey of index area conducted by DFO. A total of 807 chinook spawners was counted.

Recommendations for 2006

Participate in aerial enumeration surveys. Collect DNA samples. Check reaches above Thirtymile Creek confluence for chum spawning potential.

Tributaries:

Wolf River - 8740227

drainage area: 3,954 km²

The Wolf River enters the Nisutlin River from the east 6 km upstream from the mouth.

Chinook salmon spawn and rear in the river. Aerial surveys have been conducted by DFO and the Alaska Dept. of Fish and Game (ADF&G) to enumerate chinook since 1970. Access to the lower river is by boat from the Nisutlin River, and to the upper river is by floatplane to Wolf Lake.

2003

August 23 - Conducted aerial survey with DFO personnel to enumerate chinook spawners.

August 25-27 - Flew to Wolf Lake. Collected DNA samples from chinook spawners at lake outlet and sent to Pat Milligan (DFO).

Recommendations for 2004

Continue to participate in aerial enumeration surveys. Continue DNA sample collection.

2004

August 18 - Aerial chinook enumeration survey of index area conducted by DFO. A total of 226 chinook spawners was counted.

Recommendations for 2005

Participate in aerial enumeration surveys. Continue DNA sample collection.

2005

DNA sample collection was deferred to accommodate other DNA collection projects in the Teslin River Watershed.

August 15 - Aerial chinook enumeration survey of index area conducted by DFO. A total of 260 chinook spawners was counted.

Recommendations for 2006

Participate in aerial enumeration surveys. Continue DNA sample collection.

Sidney Creek - 8743655129

drainage area: 800 km²

Sidney Creek enters the Nisutlin from the west approximately 63 km upstream from the mouth. Chinook salmon spawn and rear in the creek. There is placer mining in and around the upper drainage. There has been industrial forestry in the lower drainage basin. Access is by the Canol Road, which crosses the creek near the mouth. Iron Creek Road provides access to the upper creek at and above the mouth of Iron Creek.

2003

June 8 - flew creek. Saw two beaver dams 30 km upstream from Nisutlin River confluence.

July 22-23 - set minnow traps above and below Canol Road culvert. Captured jcs both above and below culvert.

August 29 - conducted ground survey of creek. No chinook spawners seen at this time. One adult chinook observed last week by Howard Smith (Teslin Habitat Steward) near the Canol Road culvert.

Recommendations for 2004

Check stream for beaver activity. Monitor spawning activity. Monitor jcs utilisation.

2004

June 20 - survey creek from mouth to approximately 1.3 km upstream. No obstructions observed.

July 29 - set 4 minnow traps below highway culvert (captured 401 jcs). Set 5 minnow traps above highway culvert (captured 197 jcs (two jcs were 1+)). Sampled subset of total jcs captured.

Recommendations for 2005

Check stream for beaver activity. Monitor spawning activity. Monitor jcs utilisation.

2005

No survey conducted due to time constraints.

Recommendations for 2006

Effort should be made to check stream for beaver activity and monitor utilisation by adult and juvenile chinook salmon.

Tributaries:

Iron Creek - 8743655129646

drainage area:

Iron Creek enters Sidney Creek from the north approximately 23 km upstream from the mouth. A waterfall near the mouth obstructs all access to the upper creek. Access is from the Iron Creek road.

2003

July 22 - unable to drive to Iron Creek because of poor road conditions.

Recommendations for 2004

Check for jcs utilisation downstream of falls if road is accessible.

2004

No survey conducted.

Recommendations for 2005

Check for jcs utilisation downstream of falls if road is accessible.

2005

No survey conducted due to time constraints.

Recommendations for 2006

Effort should be made to check for jcs utilisation downstream of falls if road is accessible.

Evelyn Creek - 8743655129116

drainage area: 210 km²

Evelyn Creek enters Sidney Creek from the south approximately 7 km from the mouth. Chinook salmon rear in the creek and there is anecdotal information of salmon spawning. Placer mining occurs in the drainage. Access is from the Canol Road crossing near the mouth. Evelyn Creek road provides access to the upper reaches.

2003

June 8 - flew creek. No active beaver dams observed.

July 21-22 - set minnow traps above and below Canol road bridge. No jcs captured.

Recommendations for 2004

Check stream for beaver activity. Monitor jcs utilisation.

2004

June 16 - survey creek 400 meters d/s of highway bridge and 250 meters above bridge.

No obstructions observed.

Recommendations for 2005

Check stream for beaver activity. Monitor jcs utilisation.

2005

No survey conducted due to time constraints.

Recommendations for 2006

Effort should be made to check stream for beaver activity and to monitor jcs utilisation.

Murphy Creek - 8743655

drainage area: 210 km²

Murphy Creek enters Sidney Creek from the south approximately 4 km upstream from the mouth. Chinook salmon rear in the creek. Chinook spawning has not been documented but habitat may be suitable. Access is from the Canol Road crossing. This culvert is a barrier to fish passage during low flows.

2003

June 8 - flew creek. no beaver dams observed.

July 21-22 - set minnow traps above and below culvert. No jcs captured at this time.

Recommendations for 2004

Check stream for beaver activity. Monitor jcs utilisation above and below highway culvert.

2004

July 29 - conducted culvert assessment.

Recommendations for 2005

Check stream for beaver activity. Monitor jcs utilisation above and below highway culvert.

2005

No survey conducted due to time constraints.

Recommendations for 2006

Effort should be made to check stream for beaver activity and monitor jcs utilisation.

Sidney Lake Outlet Creek - 874395 (Sidney Lake)

drainage area:

This creek enters the Nisutlin River approximately 68 km upstream from the mouth. Nothing is known about chinook utilisation in this creek. Access is by boat from Sidney Lake and Nisutlin River. Beavers may dam the creek.

2003

June 26 - took boat from Sidney Lake. Numerous log jams and old beaver dams seen throughout the creek.

Recommendations for 2004

Check for jcs utilisation

2004

June 20 - accessed mouth of creek by boat from Nisutlin River. Surveyed lower 600 meters of creek. No obstructions observed.

Recommendations for 2005

Check for jcs utilisation.

2005

No survey conducted due to time constraints.

Recommendations for 2006

Check for jcs utilisation if time permits.

Cottonwood Creek - 874437

drainage area: 75.6 km²

Cottonwood Creek enters the Nisutlin River from the west approximately 76 km upstream from the mouth. Chinook salmon rear in the creek. Access is by the Canol Road crossing near the mouth. Beavers may dam the creek. The highway culvert is a barrier for fish passage.

2003

June 26 - ground survey of creek. Saw beaver dam downstream of culvert.

July 23-24 - set minnow traps above and below culvert. Captured jcs below culvert and above beaver dam. No jcs captured above culvert. Good rearing habitat observed upstream of culvert. Culvert hung 1.5 meters above creek level. Took photos.

August 12 - ground survey for chinook spawner utilisation. No spawners observed.

Recommendations for 2004

Check stream for beaver activity. Monitor utilisation by jcs.

2004

June 15 - set 5 minnow traps below highway culvert (no jcs captured). Set 4 minnow traps above highway culvert (no jcs captured).

July 21 - set 4 minnow traps below highway culvert (captured 167 jcs). Sampled subset of jcs. Set 5 minnow traps above highway culvert (no jcs captured). Conducted culvert assessment.

Recommendations for 2005

Initiate culvert passage mitigation. Check stream for beaver activity. Monitor utilisation by jcs.

2005

The process to mitigate culvert passage problem was not initiated in 2005.

July 7 - surveyed creek 200 m upstream and 200 m downstream of highway culvert. No beaver dams observed. Set 5 minnow traps above highway culvert (no jcs captured). Set 5 minnow traps below highway culvert (no jcs captured).

Recommendations for 2006

Initiate culvert passage mitigation. Check stream for beaver activity. Monitor utilisation by jcs.

Coyote Creek -

drainage area:

Coyote Creek enters the Nisutlin River from the west approximately 73 km upstream from the mouth. Chinook salmon rear in the creek. Access is from the Canol Road crossing. Beaver may dam the creek.

2003

July 23- 24 - set minnow traps above and below culvert. Captured jcs both above and below culvert. Saw lots of fresh beaver sign. Removed beaver sticks accumulated above culvert.

Took photos. Saw black bear.

Recommendations for 2004

Check stream for beaver activity. Monitor jcs utilisation.

2004

July 22 - Removed logjam d/s of highway culvert. Observed scouring and erosion of stream banks below culvert. Conducted culvert assessment.

July 27 - set 5 minnow traps below highway culvert (no jcs captured). Set 4 minnow traps above highway culvert (no jcs captured) - possible barrier downstream not investigated.

Recommendations for 2005

Check stream for beaver activity. Monitor jcs utilisation. Initiate measures to stabilize stream banks at culvert outfall.

2005

No survey conducted due to time constraints. The stabilisation of stream banks at the culvert outfall was not given priority in the 2005 work plan.

Recommendations for 2006

Effort should be made to check stream for beaver activity and monitor jcs utilisation. Re-investigate stream to determine the extent of bank destabilisation. Options for stabilising banks should be explored if necessary.

MORLEY RIVER

Morley River – 875

drainage area:

The Morley River enters Morley Bay approximately 16 km south of the town of Teslin. The lower river extends from the mouth to Morley Lake, and the upper river extends from Morley Lake to the outlet of Morris Lake. Chinook salmon spawn and rear in the lower river. Less is known of the upper river, with questionable accounts of spawning upstream as far as Morris Lake. The Alaska Highway follows the river to Morley Lake and provides multiple accesses to the river. Access can also be gained by boat from Teslin Lake and Morley Lake; however,

there is a set of falls between the lakes. There is potential for forestry and hydro development in the watershed.

2003

August 15 - Collected DNA samples from chinook spawners and sent to Pat Milligan (DFO).

Recommendations for 2004

continue DNA sample collection. Investigate spawning potential and jcs utilisation above Morley Lake.

2004

June 8 - set 5 minnow traps between highway bridge and Morley Lake (captured one over-wintering jcs (1+)).

August 13 & 16 - angled for adult salmon to collect DNA samples.

August 17 - assisted DFO personnel with egg-take for 'Salmon in the Classroom' program. Collected DNA samples from adult salmon.

August 19 - set 1 minnow trap at outlet of Morley Lake (captured 1 jcs). Sampled jcs captured. Set 1 minnow trap at inlet of Morley Lake (no jcs captured). Set minnow traps and beach-seined shoreline of Morley Lake to demonstrate fish collection techniques to children participating in the Teslin community recreational 'Youth Environment camp'. Captured juvenile whitefish, burbot, northern pike and sculpins (no jcs captured).

Recommendations for 2005

continue DNA sample collection. Investigate spawning potential and jcs utilisation above Morley Lake. Collect baseline information on habitat and salmon utilisation in tributaries to Morley Lake.

2005

DNA sample collection was deferred to accommodate other DNA collection projects in the Teslin River Watershed.

July 25 – conducted stream assessments on two unnamed tributaries of Morley Lake. Set minnow traps in both tributaries. No jcs captured.

Recommendations for 2006

Continue DNA sample collection in Morley River. Investigate spawning potential and jcs utilisation above Morley Lake.

Tributaries:

Hays Creek – 8750541

drainage area:

Hays Creek enters the Morley River from the north approximately 4.6 km from the mouth. Chinook salmon rear in the stream. The culvert is a velocity barrier to juvenile fish. Reaches above the culvert have good rearing potential. Access is from the Alaska Highway crossing.

2003

July 17 - Set minnow traps above and below culvert. Juvenile chinook captured only below culvert. High water velocity observed through culvert.

Recommendations for 2004

Investigate methods to mitigate culvert passage problem. Monitor jcs utilisation.

2004

no survey conducted.

Recommendations for 2005

Investigate methods to mitigate culvert passage problem. Monitor jcs utilisation.

2005

no survey conducted due to time constraints.

Recommendations for 2006

Conduct stream habitat assessment above highway culvert to determine extent of rearing potential. Consider moving some jcs from below culvert to reaches above if habitat is suitable for rearing.

Strawberry Creek - 8750909

drainage area:

Strawberry creek enters the Morley River approximately 9 km upstream from the mouth. Chinook salmon rear in the creek. Access is from the Alaska Highway crossing.

2004

June 21 - survey creek on foot. No obstructions observed downstream to mouth and immediately upstream of highway culvert. Culvert does not appear to be a barrier but this was not verified.

Recommendations for 2005

check stream for jcs utilisation.

2005

No survey conducted due to time constraints.

Recommendations for 2006

Effort should be made to check stream for jcs utilisation.

Lower Hazel Creek - 8407132

drainage area:

Lower Hazel Creek enters Morley Lake from the south approximately 10 km from the outlet. Little is known of this small stream. Access is from the Alaska Highway crossing near the mouth. Beavers may dam the creek.

2003

July 3 - Survey creek on foot. Saw 3 active beaver dams. Took photos. Culvert appears passable by fish.

Recommendations for 2004

check stream for jcs utilisation. Check stream for beaver activity.

2004

June 9 - beaver dam assessment training with DFO personnel. Observed series of four active beavers dams upstream of highway culvert approximately 50 meters upstream of highway culvert.

Recommendations for 2005

check stream for jcs utilisation and remove obstructions to maintain upstream migration for jcs if necessary.

2005

no survey conducted due to time constraints.

Recommendations for 2006

Effort should be made to check stream for jcs utilisation.

GLADYS RIVER

Gladys River - 817

Gladys River enters Teslin Lake from the west. Chinook salmon spawn and rear in the river, primarily in the lower section of the river downstream of Hall Lakes. A falls approximately 6 kms upstream of the mouth is negotiable by adult chinook but would be a barrier to upstream migrating jcs. Access to the lower river is by boat from Teslin Lake, and to the upper river is by floatplane to one of the lakes.

2003

August 13 - Went by boat to mouth of river to collect DNA samples. No adult chinook observed.

August 18-20 - flew into Hall Lakes. Collected DNA samples and sent to Pat Milligan (DFO). Saw 10 chinook spawners at the outlet. Set five minnow traps overnight. Captured jcs downstream of lake outlet.

Recommendations for 2004

continue DNA sample collection.

2004

no survey conducted

Recommendations for 2005

continue DNA sample collection.

2005

DNA sample collection was deferred to accommodate other DNA collection projects in the Teslin River Watershed.

Recommendations for 2006

Collect DNA samples.

SWIFT RIVER

(note: there is another Swift River, which enters the Lower Teslin River)

Swift River - 878

drainage area: 4,202 km²

The Swift River enters Teslin Lake from the east approximately 57 km from the town of Teslin. Chinook salmon spawn and rear in the river. The Alaska Highway follows the river to the headwaters and provides multiple accesses. Access to the lower reaches can be gained by boat from Teslin Lake, and to the upper reaches from Swan Lake. The section of river below Swan Lake is a popular corridor for boaters particularly during the moose hunting season.

2003

August 13 - Went by boat to mouth of river to collect DNA samples. No adult chinook observed.

August 21-22 - surveyed areas accessible from highway. Attempted to collect DNA samples but no chinook observed.

Recommendations for 2004

Continue DNA sample collection

2004

no survey conducted due to extensive forest fires in the drainage.

Recommendations for 2005

Continue DNA sample collection. Conduct fisheries assessments in tributaries of the Swift River upstream of Swan Lake to determine utilisation by jcs. Determine extent of utilisation upstream of Swan Lake by chinook salmon spawners.

2005

Fisheries assessments were deferred to accommodate other projects

Recommendations for 2006

Conduct fisheries assessments in tributaries of the Swift River upstream of Swan Lake to determine utilisation by jcs. Determine extent of utilisation upstream of Swan Lake by chinook salmon spawners. Collect DNA samples if feasible.

Tributaries:

Smart River - 8783927

drainage area:

The Smart River enters Swift River from the north approximately 40 km upstream from the mouth. Chinook salmon spawn and rear in the river, but little is known of utilisation in the upper reaches. Mineral exploration occurs in the upper drainage. Access to the lower reaches is from the Alaska Highway crossing at the mouth. The upper reaches can be accessed from mining roads, which follow both sides of the creek.

2003

August 28 - surveyed area above highway bridge on foot. Saw no sign of beaver activity.

Observed adult grayling and whitefish species.

Attempted to collect DNA samples but no chinook observed.

Recommendations for 2004

Collect DNA samples. Conduct aerial survey to locate spawning areas.

2005

fisheries assessments were deferred to accommodate other projects.

Recommendations for 2006

Conduct aerial survey to locate spawning areas and collect DNA samples if feasible.

Log jam Creek – 8784206

drainage area:

Logjam creek enters the Swift River from the north approximately 53.5 km from the mouth. Chinook salmon rear in the creek. Salmon utilisation of the upper drainage basin is obstructed by falls. Access is from the Alaska Highway crossing.

2003

June 24 - Survey creek from mouth to highway bridge on foot. No beaver dams observed.

Waterfalls deny access to the upper watershed.

Recommendations for 2004

check stream for jcs utilisation.

2004

June 18 - surveyed creek on foot. No obstructions observed downstream to mouth and immediately upstream of highway culvert.

Recommendations for 2005

check stream for jcs utilisation.

2005

no survey conducted due to time constraints

Recommendations for 2006

Effort should be made to check stream for jcs utilisation.

YUKON RIVER MID-MAINSTEM WATERSHED

(The Teslin Tlingit Council Traditional Territory includes much of the Big Salmon River drainage)

BIG SALMON RIVER

Big Salmon River - 860

drainage area: 6,760 km²

The Big Salmon River is the outlet stream of the Quiet Lake/Big Salmon Lake chain.

Chinook salmon spawn and rear in the river. Aerial enumeration surveys of an index area have been conducted by DFO and ADF&G since 1970. The river can be accessed by boat from Quiet Lake along the Canol Road or from the Yukon River along the Robert Campbell and the Klondike Highways near Carmacks or from Whitehorse.

2003

August 23 - Conducted aerial survey with DFO personnel to enumerate chinook spawners in index area.

Recommendations for 2004

continue to participate in aerial enumeration surveys. Collect baseline information on habitat and salmon utilisation in tributaries to Quiet Lake.

2004

August 18 - Aerial chinook enumeration survey of index area conducted by DFO. A total of 762 chinook spawners was counted.

Recommendations for 2005

participate in aerial enumeration surveys.

2005

August 16-18 - Collected tissue samples for DNA analysis and age-sex-length data from post-spawned chinook salmon with assistance from DFO personnel. Samples were collected downstream of Big Salmon Lake by angling. Access was by ATV to boat launch on Big Salmon Lake and by inflatable boat to collection site. Collected 17 samples.

August 15 - Aerial chinook enumeration survey of index area conducted by DFO. A total of 952 chinook spawners was counted. The total run size determined from sonar operations at the mouth of the river was 5,618 chinook salmon.

Recommendations for 2006

Participate in aerial enumeration surveys. Collect DNA samples.

Quiet Lake - 860

drainage area:

Quiet Lake is a headwater lake of the Big Salmon River. The lake is 31 km in length, has an average width of 1.7 km and an average depth of 32.9 m. Access is from the South Canol Road following the east shore. Boats can be launched at the YTG campground.

2004

August 10 - beach seined along lakeshore. Captured one jcs from 2 sets. Other fish species captured included juvenile whitefish (41) and slimy sculpins. Set minnow traps at lake outlet; captured and sampled one jcs.

Recommendations for 2005

Collect baseline information on habitat and salmon utilisation in tributaries to Quiet Lake.

2005

August 1-3 - conducted stream assessments of tributaries to Quiet and Sandy Lake.

Tributaries investigated included: the inlet creek (unnamed) and Crater Creek flowing into Quiet Lake; and Brown Creek flowing into Sandy Lake. Minnow traps were set in all 4 tributaries. No jcs captured.

Recommendations for 2006

Collect baseline information on habitat and salmon utilisation in tributaries of the upper reaches of the Big Salmon River.

Appendix 3. **Obstruction Removal and Maintenance of Upstream Migration for
Salmon Field Sheets**

Dave Creek	27
Hundred Mile Creek.....	27
Swift River (north).....	28
1st Beaver Creek	28
2nd Beaver Creek	29
Brooks Brook.....	29
Brooks Brook – 2nd assessment	30
Brooks Brook – 3rd assessment	30
Brooks Brook – 4th assessment.....	31
Sterlin Creek.....	31
Sterlin Creek – 2nd assessment.....	32
Grayling Creek.....	32
Grayling Creek – 2nd assessment	33
Cottonwood Creek	33
Deadman Creek.....	34

Dave Creek

Stream/River Name: Dave Creek	Date: July 20, 2005
Crew: Darren, Joe, Frank, George	R&E Fund Project #: 47-05
Type of Activity: ☒ assessment ☒ obstruction removal ☐ monitoring	
Type of Obstruction: ☒ Beaver Dam ☐ Log jam ☐ other (specify)	
Location of Obstruction (describe or number): at the mouth of Dave Creek	
UTM Northing (or latitude): 08V057430 UTM Easting (or longitude): 6728650	
Fish above obstruction (captured/observed): ☐ yes ☒ no List fish species captured/observed*: (specify adults (A) &/or juveniles (J))	
Fish below obstruction (captured/observed): ☐ yes ☒ no List fish species captured/observed*: (specify adults (A) &/or juveniles (J))	
COMMENTS/description of activities: walked creek upstream 200 m. good juvenile fish habitat observed. Breached beaver dam. No G-traps set	

Hundred Mile Creek

Stream/River Name: Hundred Mile Creek	Date: July 20,2005
Crew: Darren, Joe, Frank, George	R&E Fund Project #: 47-05
Type of Activity: ☒ assessment ☐ obstruction removal ☐ monitoring	
Type of Obstruction: ☐ Beaver Dam ☐ Log jam ☐ other (specify)	
Location of Obstruction (describe or number):	
UTM Northing (or latitude): UTM Easting (or longitude):	
Fish above obstruction (captured/observed): ☐ yes ☐ no List fish species captured/observed*: (specify adults (A) &/or juveniles (J))	
Fish below obstruction (captured/observed): ☐ yes ☐ no List fish species captured/observed*: (specify adults (A) &/or juveniles (J))	
COMMENTS/description of activities: walked creek upstream 200m. No obstructions observed. Set 5 minnow traps, captured 12 jcs	

Swift River (north)

Stream/River Name: Swift River (north)	Date: July 20, 2005
Crew: Frank, George, Darren	R&E Fund Project #: 47-05
Type of Activity: assessment obstruction removal monitoring	
Type of Obstruction: Beaver Dam Log jam other (<i>specify</i>)	
Location of Obstruction (describe or number): 10 m upstream of mouth	
UTM Northing (or latitude): 0562179 UTM Easting (or longitude): 674220	
Fish above obstruction (captured/observed): yes no List fish species captured/observed*: (<i>specify adults (A) &/or juveniles (J)</i>) Unidentified juvenile fish observed	
Fish below obstruction (captured/observed): yes no List fish species captured/observed*: (<i>specify adults (A) &/or juveniles (J)</i>) Unidentified juvenile fish observed	
COMMENTS/description of activities: took apart some of the logjam. Set 5 minnow traps , captured 8 jcs, 3 Slimy Sculpins and 1 burbot in vicinity of obstruction.	

1st Beaver Creek

Stream/River Name: 1 st Beaver Creek	Date: July 27, 2005
Crew: Frank, Darren, George	R&E Fund Project #: 47-05
Type of Activity: assessment obstruction removal monitoring	
Type of Obstruction: Beaver Dam Log jam other (<i>specify</i>)	
Location of Obstruction (describe or number):	
UTM Northing (or latitude): UTM Easting (or longitude):	
Fish above obstruction (captured/observed): yes no List fish species captured/observed*: (<i>specify adults (A) &/or juveniles (J)</i>)	
Fish below obstruction (captured/observed): yes no List fish species captured/observed*: (<i>specify adults (A) &/or juveniles (J)</i>)	
COMMENTS/description of activities: no obstructions, lots of unidentified fish, no G-traps set	

2nd Beaver Creek

Stream/River Name: 2 nd Beaver Creek	Date: July 27, 2005
Crew: George, Darren, Frank	R&E Fund Project #: 47-05
Type of Activity: ☒ assessment ☐ obstruction removal ☐ monitoring	
Type of Obstruction: ☐ Beaver Dam ☐ Log jam ☐ other (<i>specify</i>)	
Location of Obstruction (describe or number):	
UTM Northing (or latitude): UTM Easting (or longitude):	
Fish above obstruction (captured/observed): ☐ yes ☐ no List fish species captured/observed*: (<i>specify adults (A) &/or juveniles (J)</i>)	
Fish below obstruction (captured/observed): ☐ yes ☐ no List fish species captured/observed*: (<i>specify adults (A) &/or juveniles (J)</i>)	
COMMENTS/description of activities: walked up creek about 500 m saw old washed out beaver dam, no obstructions, no G-traps set	

Brooks Brook

Stream/River Name: Brooks Brook	Date: July 12, 2005
Crew: Darren, George, Frank	R&E Fund Project #: 47-05
Type of Activity: ☒ assessment ☐ obstruction removal ☐ monitoring	
Type of Obstruction: ☒ Beaver Dam ☐ Log jam ☐ other (<i>specify</i>)	
Location of Obstruction (describe or number): 1 st dam 200 m below culvert, old beaver dam 250 m below	
UTM Northing (or latitude): 6699012 UTM Easting (or longitude): 0599446	
Fish above obstruction (captured/observed): ☐ yes ☒ no List fish species captured/observed*: (<i>specify adults (A) &/or juveniles (J)</i>)	
Fish below obstruction (captured/observed): ☐ yes ☒ no List fish species captured/observed*: (<i>specify adults (A) &/or juveniles (J)</i>)	
COMMENTS/description of activities: set 5 minnow traps below dam and 6 minnow traps above dam,	

Brooks Brook – 2nd assessment

Stream/River Name: Brooks Brook	Date: July 13, 2005
Crew: Darren, George, Frank	R&E Fund Project #: 47-05
Type of Activity: assessment obstruction removal monitoring	
Type of Obstruction: Beaver Dam Log jam other (<i>specify</i>)	
Location of Obstruction (describe or number): 1 st dam 200 m below culvert, old inactive beaver dam 250 meters below culvert	
UTM Northing (or latitude): 6699012	UTM Easting (or longitude): 0599446
Fish above obstruction (captured/observed): yes no List fish species captured/observed*: (<i>specify adults (A) &/or juveniles (J)</i>) captured 1 jcs , 3 SS	
Fish below obstruction (captured/observed): yes no List fish species captured/observed*: (<i>specify adults (A) &/or juveniles (J)</i>) captured 14 jcs, SS, 1 BB	
COMMENTS/description of activities: breached dams	

Brooks Brook – 3rd assessment

Stream/River Name: Brooks Brook	Date: July 14, 2005
Crew: Darren, George, Frank	R&E Fund Project #: 47-05
Type of Activity: assessment obstruction removal monitoring	
Type of Obstruction: Beaver Dam Log jam other (<i>specify</i>)	
Location of Obstruction (describe or number): 1 st dam 200 m below culvert, old inactive beaver dam 250 meters below culvert	
UTM Northing (or latitude): 6699012	UTM Easting (or longitude): 0599446
Fish above obstruction (captured/observed): yes no List fish species captured/observed*: (<i>specify adults (A) &/or juveniles (J)</i>) saw unidentified juvenile fish	
Fish below obstruction (captured/observed): yes no List fish species captured/observed*: (<i>specify adults (A) &/or juveniles (J)</i>) saw unidentified juvenile fish	
COMMENTS/description of activities: dams had remained open. No G-traps set	

Brooks Brook – 4th assessment

Stream/River Name: Brooks Brook	Date: August 9-10, 2005
Crew: Darren, George, Frank	R&E Fund Project #: 47-05
Type of Activity: assessment obstruction removal monitoring	
Type of Obstruction: Beaver Dam Log jam other (<i>specify</i>)	
Location of Obstruction (describe or number): 1 st dam 200 m below culvert, old inactive beaver dam 250 meters below culvert	
UTM Northing (or latitude): 6699012	UTM Easting (or longitude): 0599446
Fish above obstruction (captured/observed): yes no List fish species captured/observed*: (<i>specify adults (A) &/or juveniles (J)</i>)	
Fish below obstruction (captured/observed): yes no List fish species captured/observed*: (<i>specify adults (A) &/or juveniles (J)</i>) captured 8 jcs	
COMMENTS/description of activities: beaver dams remain open, set 8 traps 300 m upstream of mouth to culvert	

Sterlin Creek

Stream/River Name: Sterlin Creek	Date: July 4, 2005
Crew: George, Frank, John, Darren	R&E Fund Project #: 47-05
Type of Activity: assessment obstruction removal monitoring	
Type of Obstruction: Beaver Dam Log jam other (<i>specify</i>)	
Location of Obstruction (describe or number): 100 m up Sterling Creek from mouth	
UTM Northing (or latitude): 6673542	UTM Easting (or longitude): 0615896
Fish above obstruction (captured/observed): yes no List fish species captured/observed*: (<i>specify adults (A) &/or juveniles (J)</i>) Observed adult grayling	
Fish below obstruction (captured/observed): yes no List fish species captured/observed*: (<i>specify adults (A) &/or juveniles (J)</i>) Observed adult grayling and unidentified juvenile fish	
COMMENTS/description of activities: took apart some of the dam	

Sterlin Creek – 2nd assessment

Stream/River Name: Sterlin Creek	Date: Aug 11, 2005
Crew: DS, FS, GW	R&E Fund Project #: 47-05
Type of Activity: ☒ assessment ☒ obstruction removal ☐ monitoring	
Type of Obstruction: ☒ Beaver Dam ☐ Log jam ☐ other (<i>specify</i>)	
Location of Obstruction (describe or number): 200 m up Sterling	
UTM Northing (or latitude):	UTM Easting (or longitude):
Fish above obstruction (captured/observed): ☒ yes ☐ no List fish species captured/observed*: (<i>specify adults (A) &/or juveniles (J)</i>) Adult grayling	
Fish below obstruction (captured/observed): ☒ yes ☐ no List fish species captured/observed*: (<i>specify adults (A) &/or juveniles (J)</i>) Adult grayling, unidentified juvenile fish	
COMMENTS/description of activities: dam 100 m upstream remained open, beaver dam 200 m upstream very high. Took apart some of the dam	

Grayling Creek

Stream/River Name: Grayling Creek	Date: July 27, 2005
Crew: George, Darren, Frank	R&E Fund Project #: 47-05
Type of Activity: ☐ assessment ☒ obstruction removal ☐ monitoring	
Type of Obstruction: ☒ Beaver Dam ☐ Log jam ☐ other (<i>specify</i>)	
Location of Obstruction (describe or number): 250 m up creek from mouth	
UTM Northing (or latitude):	UTM Easting (or longitude):
Fish above obstruction (captured/observed): ☐ yes ☒ no List fish species captured/observed*: (<i>specify adults (A) &/or juveniles (J)</i>)	
Fish below obstruction (captured/observed): ☒ yes ☐ no List fish species captured/observed*: (<i>specify adults (A) &/or juveniles (J)</i>) Unidentified fish observed	
COMMENTS/description of activities: assessment of beaver dam. Very big beaver dam about 8' beaver still active, took apart some of the dam	

Grayling Creek – 2nd assessment

Stream/River Name: Grayling Creek	Date: August 11, 2005
Crew: DS, FS, GW	R&E Fund Project #: 47-05
Type of Activity: assessment obstruction removal monitoring	
Type of Obstruction: Beaver Dam Log jam other (specify)	
Location of Obstruction (describe or number): 250 m up creek	
UTM Northing (or latitude):	UTM Easting (or longitude):
Fish above obstruction (captured/observed): yes no List fish species captured/observed*: (specify adults (A) &/or juveniles (J))	
Fish below obstruction (captured/observed): yes no List fish species captured/observed*: (specify adults (A) &/or juveniles (J))	
COMMENTS/description of activities: beaver dam built up. No action taken. Below dam water level too low, dam too high for fish to get above, beaver still active.	

Cottonwood Creek

Stream/River Name: Cottonwood Creek	Date: July 6, 2005
Crew: Darren, Frank, George	R&E Fund Project #: 47-05
Type of Activity: assessment obstruction removal monitoring	
Type of Obstruction: Beaver Dam Log jam other (specify) culvert	
Location of Obstruction (describe or number): at highway	
UTM Northing (or latitude):	UTM Easting (or longitude):
Fish above obstruction (captured/observed): yes no List fish species captured/observed*: (specify adults (A) &/or juveniles (J)) Slimy sculpins	
Fish below obstruction (captured/observed): yes no List fish species captured/observed*: (specify adults (A) &/or juveniles (J))	
COMMENTS/description of activities: set 5 traps 200 up from culvert, set 5 traps 200 m down from culvert. No juvenile chinook captured. Water turbid, small beaver dam above culvert	

Deadman Creek

Stream/River Name: Deadman Creek	Date: July 6, 2005
Crew: Darren, Frank, George	R&E Fund Project #: 47-05
Type of Activity: ☒ assessment ☐ obstruction removal ☐ monitoring	
Type of Obstruction: ☐ Beaver Dam ☐ Log jam ☒ other (specify) culvert	
Location of Obstruction (describe or number): at highway	
UTM Northing (or latitude): UTM Easting (or longitude):	
Fish above obstruction (captured/observed): ☒ yes ☐ no List fish species captured/observed*: (specify adults (A) &/or juveniles (J)) Slimy sculpins	
Fish below obstruction (captured/observed): ☐ yes ☒ no List fish species captured/observed*: (specify adults (A) &/or juveniles (J))	
COMMENTS/description of activities: set 5 traps 200 up from culvert, set 5 traps 200 m down from culvert. No juvenile chinook captured. Water turbid, small beaver dam above culvert	

Appendix 4. **Stream Habitat and Fisheries Assessment Field Sheets, 2005**

Quiet Lake Tributary – Crater Creek	36
Quiet Lake Tributary – Inlet Stream (unnamed).....	37
Sandy Lake Tributary – Brown Creek	38
Morley Lake Tributary – Unnamed tributary on north shore	39
Morley Lake Tributary – Unnamed tributary on south shore	40

Quiet Lake Tributary – Crater Creek

Stream Survey Data, 2005		QUIET LAKE TRIBUTARIES					
Waterbody ID	Crater Ck	Weather	Overcast, cool		Date: Aug 2, 2005		Crew: JW,DS,FS
Site ID	R#1 S#1	Northing	6771529		Easting		0603450
STREAM CHARACTERISTICS			Distance Surveyed (m)	300 m	Photos	Yes	
Temp.(°C)	7.4	Habitat Type (%)	Substrate Type (%)		Cover Type (%)		
Cond. (uS/cm)	70	Pool	20 %	Organic		Deep pool	50 %
pH	7.5	Run	50 %	Silt	10 %	LOD	10 %
Colour	Clear	Riffle	30 %	Sand		Boulder	
Stage (H/M/L)	M	Flat		Gravel	20 %	Cutbank	10 %
Gradient (%)	1%	Rapids		Cobble	70 %	Instream veg.	
Wet Width (m)	8.4	Other		Boulder		Overstream veg.	30 %
Chnl Width (m)	11.2			Bedrock			
Ave. Depth (m)	0.65	Stream Bank		Substrate Type (%)			
Vel'y (m/s)	0.33	Slope (%)	10 %	Organic		Cobble	70 %
		Height (m)	1.5	Silt	10 %	Boulder	
		% unstable	40 %	Sand	10 %	Bedrock	
				Gravel	10 %	Other	
Riparian Veg: spruce, alder, willow							
Comments: large accumulation of LWD on gravel bar in lower part of creek (not a barrier)							

Stream Survey Data, 2005			QUIET LAKE		Stream body: Crater Creek		
FISHERIES RESOURCES				Species Composition & Abundance			
Sampling Methods			Species	YOY	Juvenile	Adult	Total
Method	Hrs or min. set	# of sets	Chinook				
MT	24 hrs	7	Lake Trout				
BS			Bull Trout				
GN			Round Whitefish				
AN			Lake Whitefish				
EF			Broad Whitefish				
COMMENTS: No jcs captured			Slimy Sculpin			3	3
			Longnose Sucker				
			Arctic Grayling				
			Inconnu				
			Least Cisco				
			Northern Pike				
			Burbot				

Quiet Lake Tributary – Inlet Stream (unnamed)

Stream Survey Data, 2005		QUIET LAKE TRIBUTARIES					
Waterbody ID Site ID	Inlet Stream	Weather	Overcast, cool		Date: Aug 2, 2005		Crew: FS,DS,JW
	R#2, S#1	Northing	6759130		Easting		0605676
STREAM CHARACTERISTICS			Distance Surveyed (m)		250 m		Photos Yes
Temp.(°C)	7.0	Habitat Type (%)		Substrate Type (%)		Cover Type (%)	
Cond. (uS/cm)	130	Pool	40%	Organic		Deep pool	50%
pH	7.5	Run	60%	Silt	40%	LOD	
Colour	Green/brwn	Riffle		Sand	50%	Boulder	
Stage (H/M/L)	M	Flat		Gravel	10%	Cutbank	
Gradient (%)	<1%	Rapids		Cobble		Instream veg.	
Wet Width (m)	7.1	Other		Boulder		Overstream veg.	50%
Chnl Width (m)	7.3			Bedrock			
Depth (m)	0.62	Stream Bank		Substrate Type (%)			
Vel'y (m/s)	0.25	Slope (%)	1%	Organic	20%	Cobble	
		Height (m)	0.5	Silt	10%	Boulder	
		% unstable	30%	Sand	70%	Bedrock	
				Gravel		Other	
Riparian Veg: grasses, alder, willow							
Comments: reach is slow slough like							

Stream Survey Data, 2005			QUIET LAKE		Stream body: Inlet Stream		
FISHERIES RESOURCES			Species Composition & Abundance				
Sampling Methods			Species	YOY	Juvenile	Adult	Total
Method	Hrs or min. set	# of sets	Chinook				
MT	24 hrs	5	Lake Trout				
BS			Bull Trout				
GN			Round Whitefish				
AN			Lake Whitefish				
EF			Broad Whitefish				
			Slimy Sculpin				
COMMENTS: No fish captured			Longnose Sucker				
			Arctic Grayling				
			Inconnu				
			Least Cisco				
			Northern Pike				
			Burbot				

Sandy Lake Tributary – Brown Creek

Stream Survey Data, 2005		SANDY LAKE TRIBUTARIES					
Waterbody ID	Brown Ck	Weather	Overcast, cool		Date: Aug 3, 2005		Crew: DS,FS,JD
Site ID	R#1, S#1	Northing	6787976		Easting		0594803
STREAM CHARACTERISTICS			Distance Surveyed (m)		250 m	Photos	Yes
Temp.(°C)	7.2	Habitat Type (%)		Substrate Type (%)		Cover Type (%)	
Cond. (uS/cm)	--	Pool	10%	Organic		Deep pool	20%
pH	7.5	Run	70%	Silt	10%	LOD	10%
Colour	Clear	Riffle	20%	Sand	40%	Boulder	
Stage (H/M/L)	M	Flat		Gravel	40%	Cutbank	10%
Gradient (%)	<1%	Rapids		Cobble	10%	Instream veg.	
Wet Width (m)	9.5	Other		Boulder		Overstream veg.	60%
Chnl Width (m)	18.9			Bedrock			
Ave Depth (m)	0.85	Stream Bank		Substrate Type (%)			
Vel'y (m/s)	0.50	Slope (%)	1%	Organic		Cobble	10%
		Height (m)	0.5	Silt	10%	Boulder	
		% unstable	40%	Sand	40%	Bedrock	
				Gravel	40%	Other	
Riparian Veg: spruce, alder, willow							
Comments:							

Stream Survey Data, 2005		SANDY LAKE		Stream body: Brown Creek			
FISHERIES RESOURCES			Species Composition & Abundance				
Sampling Methods			Species	YOY	Juvenile	Adult	Total
Method	Hrs or min. set	# of sets	Chinook				
MT	23 hrs	5	Lake Trout				
BS			Bull Trout				
GN			Round Whitefish				
AN			Lake Whitefish				
EF			Broad Whitefish				
			Slimy Sculpin			2	2
COMMENTS: No jcs captured			Longnose Sucker				
			Arctic Grayling				
			Inconnu				
			Least Cisco				
			Northern Pike				
			Burbot		1		1

Morley Lake Tributary – Unnamed tributary on north shore

Stream Survey Data, 2005		MORLEY LAKE TRIBUTARIES					
Waterbody ID	Trib #1 (north)	Weather		Date: July 25, 2005	Crew: DS,FS		
Site ID	R#1	Northing	6656125	Easting	0660089		
STREAM CHARACTERISTICS		Distance Surveyed (m)	250 m	Photos			
Temp.(°C)	8.8	Habitat Type (%)	Substrate Type (%)	Cover Type (%)			
Cond. (uS/cm)		Pool	Organic	Deep pool			
PH		Run	Silt 80%	LOD 50%			
Colour	Clear	Riffle	Sand	Boulder			
Stage (H/M/L)	L	Flat	Gravel	Cutbank			
Gradient (%)		Rapids	Cobble 20%	Instream veg.			
Wet Width (m)		Other	Boulder	Overstream veg. 50%			
Chnl Width (m)			Bedrock				
Depth (m)		Stream Bank	Substrate Type (%)				
Vel'y (m/s)		Slope (%)	Organic	Cobble			
		Height (m)	Silt	Boulder			
		% unstable	Sand	Bedrock			
			Gravel	Other			
Riparian Veg: extensive willow growth, 40% crown closure							
Comments: Creek has low flow and silty bottom, lots of large woody debris in stream							

Stream Survey Data, 2005		MORLEY LAKE		Stream body: Unnamed tributary on north shore (#1)			
FISHERIES RESOURCES			Species Composition & Abundance				
Sampling Methods			Species	YOY	Juvenile	Adult	Total
Method	Hrs or min. set	# of sets	Chinook				
MT	24hrs	6	Lake Trout				
BS			Bull Trout				
GN			Round Whitefish				
AN			Lake Whitefish				
EF			Broad Whitefish				
			Slimy Sculpin				
COMMENTS: No fish captured			Longnose Sucker				
			Arctic Grayling				
			Inconnu				
			Least Cisco				
			Northern Pike				
			Burbot				

Morley Lake Tributary – Unnamed tributary on south shore

Stream Survey Data, 2005		MORLEY LAKE TRIBUTARIES					
Waterbody ID Site ID	Trib #2 (south)	Weather			Date: July 25, 2005	Crew: DS,FS	
	R#1	Northing	6657456		Easting	0662935	
STREAM CHARACTERISTICS			Distance Surveyed (m)		300 m	Photos	
Temp.(°C)	9.3	Habitat Type (%)		Substrate Type (%)		Cover Type (%)	
Cond. (uS/cm)		Pool		Organic		Deep pool	
PH		Run		Silt		LOD	
Colour	Clear	Riffle		Sand	20%	Boulder	
Stage (H/M/L)	M	Flat		Gravel		Cutbank	
Gradient (%)		Rapids		Cobble	80%	Instream veg.	
Wet Width (m)		Other		Boulder		Overstream veg.	
Chnl Width (m)				Bedrock			
Depth (m)		Stream Bank		Substrate Type (%)			
Vel'y (m/s)		Slope (%)		Organic		Cobble	
		Height (m)		Silt		Boulder	
		% unstable		Sand		Bedrock	
				Gravel		Other	
Riparian Veg: abundant willow growth, 50% crown closure							
Comments:							

Stream Survey Data, 2005		MORLEY LAKE	Stream body: Unnamed tributary on south shore (#2)				
FISHERIES RESOURCES			Species Composition & Abundance				
Sampling Methods			Species	YOY	Juvenile	Adult	Total
Method	Hrs or min. set	# of sets	Chinook				
MT	24 hrs	6	Lake Trout				
BS			Bull Trout				
GN			Round Whitefish				
AN			Lake Whitefish				
EF			Broad Whitefish				
			Slimy Sculpin				
COMMENTS: No fish captured. Observed unidentified juvenile fish in stream			Longnose Sucker				
			Arctic Grayling				
			Inconnu				
			Least Cisco				
			Northern Pike				
			Burbot				

Appendix 5. **Juvenile Chinook Salmon Sampling Data, 2005**

Brooks Brook – Station 1	42
Brooks Brook – Station 2	43
Brooks Brook – 2 nd assessment	44
Deadman Creek	45
Deadman Creek – 2 nd assessment	46
Hundred Mile Creek	47
Swift River (north).....	48

Brooks Brook – Station 1

Stream/River name: Brooks Brook				Date: July 13, 2005				
Sampling station location (describe or number): Station#1, below beaver dams								
UTM Northing (or Latitude): 6698943				UTM Easting (or Longitude): 0599370				
Persons sampling: Darren, Frank, George								
Sampling method and effort (# of G-traps set, seines, etc.):				Collection license #				
set 5 G-minnow traps				R&E Fund Project # : RE-47-05				
If traps, date & time traps set: July 13, 1:00 p.m.				Water/air temp 7.4 C/23.8 C				
If traps, date & time traps pulled: July 14, 10:00 a.m.				Water/air temp				
Stream flow (circle one):			LOW	MEDIUM			HIGH	
Water clarity (circle one):			CLEAR	SOMEWHAT TURBID (cloudy)			TURBID	
<i>If you are measuring a subset (min. 30) of the total fish captured, please write the total number of fish captured in the Fish Species column, as well as the fish species. E.g. "300 Total CH" for 300 captured Chinook juveniles.</i>								
Fish Species	LENGTH	WEIGHT	Fish Species	LENGTH	WEIGHT	Fish Species	LENGTH	WEIGHT
CH (Chinook)	53mm	1.5 g						
CH	60	2.5						
CH	60	2.2						
CH	80	4.9						
CH	75	4.6						
CH	76	4.7						
CH	77	4.5						
CH	80	4.8						
CH	72	4.2						
CH	66	3.5						
CH	82	5.2						
CH	61	3.0						
CH	61	2.7						
CH	65	3.2						
1 BB								
6 SS								

fish species: CH - Chinook Salmon, SS - slimy sculpin, BB - burbot

Brooks Brook – Station 2

[illegible]

fish species: CH - Chinook Salmon, SS - Slimy Sculpin

Brooks Brook – 2nd assessment

[illegible]

fish species: CH - Chinook Salmon

Deadman Creek

[illegible]

fish species: CH - chinook salmon

Deadman Creek – 2nd assessment

Stream/River name: Deadman Creek				Date: August 10, 2005				
Sampling station location (describe or number): 500 m upstream of hwy bridge								
UTM Northing (or Latitude): 6690631				UTM Easting (or Longitude): 0606907				
Persons sampling: George, Darren, Frank								
Sampling method and effort (# of G-traps set, seines, etc.):				Collection license #				
10 G-traps set				R&E Fund Project # : RE- 47-05				
If traps, date & time traps set: Aug. 9, 2005 11:30 am				Water/air temp 8.7 C/27.5 C				
If traps, date & time traps pulled: Aug 10, 2005 10:00 am				Water/air temp				
Stream flow (circle one):		LOW		MEDIUM		HIGH		
Water clarity (circle one):		CLEAR		SOMEWHAT TURBID (cloudy)		TURBID		
<i>If you are measuring a subset (min. 30) of the total fish captured, please write the total number of fish captured in the Fish Species column, as well as the fish species. E.g. "300 Total CH" for 300 captured Chinook juveniles.</i>								
Fish Species	LENGTH	WEIGHT	Fish Species	LENGTH	WEIGHT	Fish Species	LENGTH	WEIGHT
CH (Chinook)	53mm	1.5 g						
CH	77	4.1						
CH	62	2.0						
CH	81	5.4						
CH	73	3.2						
CH	65	2.7						
CH	78	4.3						
CH	85	5.7						
CH	64	2.2						
CH	71	3.3						
CH	70	3.0						
CH	78	4.2						
1 BB								

fish species: CH - Chinook Salmon, BB - Burbot

Hundred Mile Creek

Stream/River name: Hundred Mile Creek				Date: July 20, 2005				
Sampling station location (describe or number): Station#1, mouth of creek to 200 m upstream								
UTM Northing (or Latitude): 6733482				UTM Easting (or Longitude): 0571415				
Persons sampling: Darren, Frank, George								
Sampling method and effort (# of G-traps set, seines, etc.):				Collection license #				
set 5 G-minnow traps				R&E Fund Project # : RE-47-05				
If traps, date & time traps set: July 20, 3:00 p.m.				Water/air temp 8.2 C/14.7 C				
If traps, date & time traps pulled: July 21, 10:50 a.m.				Water/air temp				
Stream flow (circle one):		LOW		MEDIUM		HIGH		
Water clarity (circle one):		CLEAR		SOMEWHAT TURBID (cloudy)			TURBID	
<i>If you are measuring a subset (min. 30) of the total fish captured, please write the total number of fish captured in the Fish Species column, as well as the fish species. E.g. "300 Total CH" for 300 captured Chinook juveniles.</i>								
Fish Species	LENGTH	WEIGHT	Fish Species	LENGTH	WEIGHT	Fish Species	LENGTH	WEIGHT
CH (Chinook)	53mm	1.5 g						
CH	55	1.3						
CH	53	1.4						
CH	59	1.9						
CH	59	2.0						
CH	57	1.8						
CH	67	2.5						
CH	68	3.0						
CH	55	1.6						
CH	57	1.9						
CH	55	1.6						
CH	55	1.7						
CH	73	3.4						

fish species: CH - Chinook Salmon

Swift River (north)

[illegible]

fish species: CH - Chinook Salmon, SS- Slimy Sculpin; BB - Burbot